

Evgeni Gavrilov

Senior QA Engineer, in QA since 2015 · **test automation** with AI since 2023

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Aurora: my portfolio project

I'm a senior QA engineer with over 10 years in QA (since 2015). Since 2023 I have focused on **test automation**, and I use AI every day to write, maintain, and run automated tests. Aurora shows how I do this on a full system: a small clinic app with a **FastAPI backend**, a **React web app**, and a **Python IoT agent** that watches medicine fridges. Every level of testing runs in CI, and a change can't merge until it passes. AI wrote most of the code; I decided what to test and checked every result.

Portfolio	https://github.com/evg-g/aurora
Live demo	https://evg-g.github.io/appointments-web/ (sign in: admin@aurora.test / password123)
Test report	https://evg-g.github.io/appointments-web/report/
QA toolkit	https://github.com/evg-g/qa-ai-toolkit

Test automation, every level in CI

Unit	pytest, Vitest + Testing Library	Integration	real Postgres + Redis with testcontainers
Contract	OpenAPI drift tests, oasdiff breaking-change check	API fuzzing	Hypothesis, Schemathesis
Mutation	mutmut, kill rate at least 78% (nightly)	Security	authz matrix, JWT and injection tests; Trivy, gitleaks, CodeQL
End-to-end	Playwright on Chromium and WebKit	Accessibility	axe on every page, WCAG 2.2 AA, light and dark
Visual + layout	Playwright screenshots and layout rules	Performance + load	Lighthouse, bundle budget; Locust p95 gate
Device, no hardware	simulator, fault injection, 7-day soak on a fake clock	Numbers	API 388 tests, 91% coverage; web 83 unit + 86 browser; device 241

How I work with AI

- **A test only counts after I have seen it fail.** An AI agent turns each ticket into code and tests, and every test must fail first, then pass. A test that never failed may not check anything.
- **That rule found a real bug I was not looking for:** the API could answer "saved" when the database had not saved the change. I opened a ticket, and it was fixed and proven the same way.
- **I check what the AI cannot judge.** I run tests the way a user would, review screenshots, and ask for full re-checks before a release.

Bugs the tests caught

- 43 high and critical CVEs (the Starlette library and the nginx base image), found by Trivy in CI and fixed.
- A local build could copy the device's private signing key into its image. Now a CI job fails if it does.
- The dashboard showed the wrong bookings when there were more than five. The visual test missed it; a new unit test now fails on the old code.
- Audit-log text failed the WCAG AA contrast check (4.08:1), found by the axe accessibility sweep.